

## IDEATION PHASE

### Brainstorm & Idea Prioritization Template

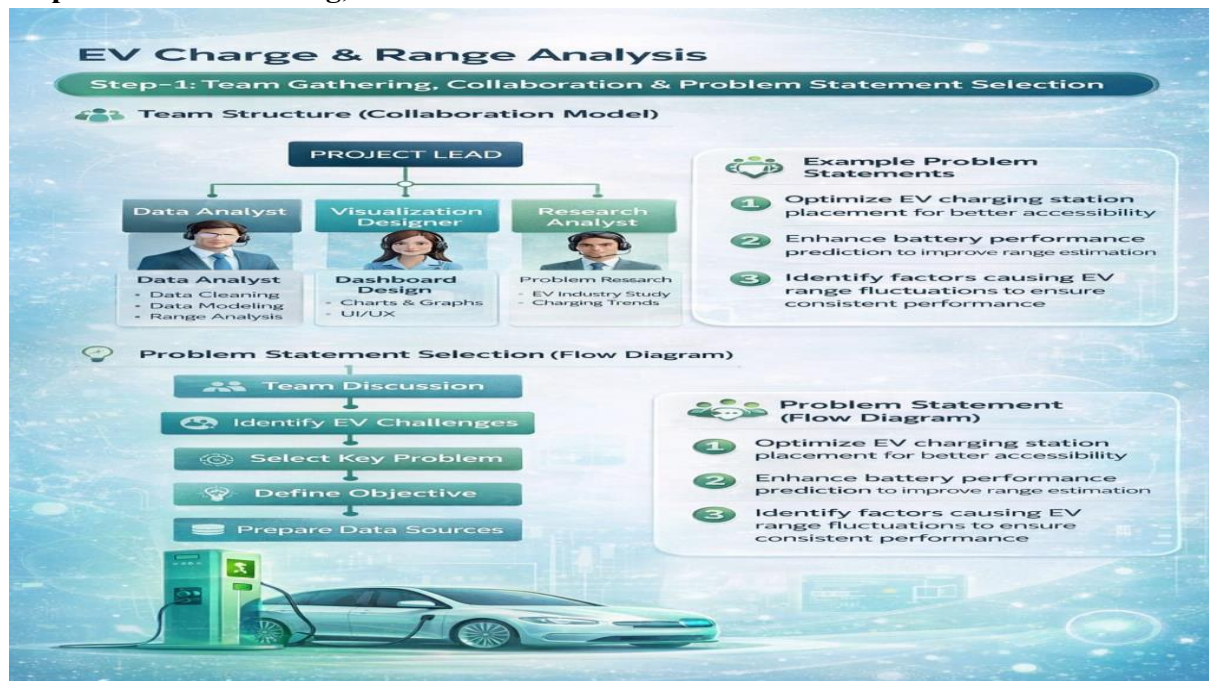
|               |                                                                     |
|---------------|---------------------------------------------------------------------|
| DATE          | 31 January 2025                                                     |
| TEAM ID       | LTVIP2025TMID30643                                                  |
| PROJECT NAME  | Visualization tools for electric vehicles charge and range analysis |
| MAXIMUM MARKS | 2 marks                                                             |

#### Brainstorm & Idea Prioritization Template:

This visualization tool is designed to help electric vehicle users clearly understand, predict, and optimize battery charge and driving range through intuitive, data-driven visuals. It transforms complex energy data—such as battery state, consumption patterns, terrain, weather, and charging behavior—into easy-to-interpret maps, charts, and interactive simulations.

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

#### Step-1: Team Gathering, Collaboration and Select the Problem Statement:



#### Step-2: Brainstorm, Idea Listing and Grouping

## Visualization Tools for Electric Vehicle (EV) Charge and Range Analysis

### Data Visualization & BI Tools



- ✓ Interactive dashboards
- ✓ Range comparison charts
- ✓ Charging station heat maps
- ✓ Battery degradation trends



### Power BI

- ✓ Real-time EV monitoring dashboards
- ✓ KPI tracking (charge %, range left, efficiency)
- ✓ Regional charging station analysis
- ✓ Predictive analytics & trends



### Step-3: Idea Prioritization

## Visualization Tools for Electric Vehicle (EV) Charge and Range Analysis

### Visualization Tools for Electric Vehicle



Charging Station Heat Maps



Battery Performance



Range Comparison